Active Maintenance Report: coppock_2014

Coppock (2014, Journal of Experimental Political Science): Information Spillovers, and its 2016 corrigendum

2026-08-01

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This repository holds the actively maintained replication code for Coppock (2014) and for the corrigendum that corrects it, together with the reproducibility report that documents what the

original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/xps.2014.9
Corrigendum	10.1017/xps.2016.8
Replication archive	10.7910/DVN/28664
Reanalyzed experiment	Butler and Nickerson (2011), 10.1561/100.00011019

The corrigendum is a separately published article with its own DOI, its own year and its own pagination, so it is cited here in its own right rather than as an attachment to the 2014 paper. It matters for what follows: the deposited archive was never updated after it appeared, so the code at the DOI above still produces the results the corrigendum retracts.

The data are not redistributed here. The deposit is 3.0 MB across 30 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published figure or table, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_2014.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 30 files, and produces every figure and table into `maintained/output/`. Required packages: tidyverse, sf, spdep, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about five minutes, most of it in the four-parameter appendix model. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

What this repository certifies is the corrigendum, not the article. The 2016 corrigendum retracts the spillover conclusion of the 2014 paper and prints corrected estimates in their place. Those are the numbers reproduced and checked here. The article's own figures are superseded by them and are not represented in the ground truth.

1.1 Does the deposited archive run?

Not as deposited, and the reason is structural rather than package rot. Every figure script opens with `load("Replication Archive/CoppockJEPS.rdata")` and writes into `Replication Archive/figures/`. No such directory is in the deposit: all 30 files sit flat at the top level, so the folder structure the code expects did not survive upload. A reader has to build three directories before any script runs. That is a five-minute repair and it is invisible until you attempt it.

Once the directories exist the figure scripts run. The data preparation script that builds their input does not: it calls `maptools`, which left CRAN in 2023.

What the archive does not contain is any trace of the correction. The deposited data still holds the corrupted similarity matrix the corrigendum describes, so the figures a reader produces from it today are the ones the corrigendum retracts.

1.2 Does the maintained rewrite reproduce the corrigendum?

Yes. 3 of the 5 corrected values return exactly, and the two that sit one grid step away do so for a reason recorded below and not because the analysis differs.

Getting there required two things that are worth stating, because either one alone leaves the corrigendum looking irreproducible.

The ideal points are not the deposited ones. Similarity between legislators is measured from W-NOMINATE scores estimated on the previous session's roll calls. The archive behind the corrigendum carries different scores from the deposited archive: 68 of the 70 differ, by up to 0.533, correlation 0.978. The transform that turns scores into a similarity matrix is byte-identical in both, `(2 - |x - y|)/2` with a zeroed diagonal, so the matrices differ only because their inputs do.

The cause is a package. Both archives call `wnominate(rollcalls, polarity = c(1, 2), minvotes = 10)`, identically. The deposited archive loads the package with no version constraint; the corrigendum archive pins it, with a comment reading "be sure to use version 0.98" and a `stopifnot` to enforce it. `wnominate` 0.98 was current when the analysis was written and 0.99 arrived in September 2014, between the analysis and publication. Running today's

`wnominate` 1.5 on the deposited roll calls puts the result closer to the corrigendum's scores (r = 0.993) than to the deposited ones (r = 0.976), which is the signature of an estimator that moved. The deposited `datapreparation.R` can no longer regenerate the deposited scores, and nothing in it says so.

The 70 scores are in `recovered/`, committed because they exist nowhere public.

The analysis predates R 3.6. The p-value surface draws 685,000 times, and R 3.6.0 replaced the algorithm behind `sample()`. Under the sampler of the day, with the corrected scores, the surface this repository produces is identical to the one saved in the corrigendum's own archive: all 1,369 cells, maximum difference exactly zero.

Those are two different questions and they are kept apart. `text_sampler_reproducibility.R` re-runs the analysis under `RNGkind(sample.kind = "Rounding")`, because whether the archive reproduces is a question about the archive. The rewrite itself keeps the current sampler, because it is the code we would write now. The two remaining differences are one grid step each and are recorded as belonging to the environment.

One published value is not identified rather than unreproduced. The confidence region reaches 0.2, the top of the grid, in the corrigendum's surface as well as this one, so the printed 0.175 is the last interior grid point rather than a bound the data support.

1.3 What this repository does not certify

The archive's figure scripts produce the article's p-value maps, and this rewrite reproduces them bit for bit where they can be compared. They are not represented in the ground truth, because the corrigendum supersedes them.

Three coding defects in the deposit are worth recording even so, since none is mentioned by the corrigendum. `CoppockJEPS_appendixfigure4.R` never calls `set.seed`, so Appendix Figure 4 differs on every run and cannot be reproduced by anyone. Five scripts read a point estimate off the first and second entries of a plateau index matrix in column-major order, which indexes the indirect axis with a position from the direct one; `figure2code.R` takes the median, which is what the paper describes. `appendixfigure5.R` draws its assignment vector once outside the simulation loop, so all fifty draws are the same assignment and every p-value is 0 or 1, which is visible in the published figure as solid blocks.

2 Paper overview

Citation: Coppock, A. (2014). "Information spillovers: Another look at experimental estimates of legislator responsiveness." *Journal of Experimental Political Science*, 1(2), 159-169. DOI: 10.1017/xps.2014.9

Corrigendum: Coppock, A. (2016). *Journal of Experimental Political Science*, 3(2), 206-208. DOI: 10.1017/xps.2016.8

Summary: Butler and Nickerson (2011) mailed district-level public opinion data to 35 of the 70 members of the New Mexico House before a 2008 vote on a tax rebate bill, in a matched-pair randomization, and estimated the effect of that information on roll call votes. The reanalysis asks what happens when the non-interference assumption is dropped, on the grounds that legislatures are information-sharing networks. Legislators are assumed to be exposed to treatment through their ideologically similar colleagues, with similarity measured from W-NOMINATE ideal points estimated on the preceding session's roll calls. Inference follows Bowers, Fredrickson and Panagopoulos (2013): each pair of hypothesized direct and indirect effects implies a uniformity trial, the hypothesis is re-imposed under draws from the 10,000 admissible randomizations, and the p-value is the share of draws whose sum of squared residuals the observed one exceeds. The article concluded that the total causal effect of treatment was about twice the difference in means. The corrigendum retracts the spillover part of that conclusion.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
CoppockJEPS_10000Randomizations.R	Runs; regenerates a matrix the deposit already carries, under a seed R has since changed	None needed: Z_block is inside the deposited CoppockJEPS.RData
CoppockJEPS_datapreparation.R	Fails: maptools is off CRAN, and it writes CoppockJEPS.rdata over the deposited CoppockJEPS.RData	sf::st_read plus spdep::poly2nb rebuilds the one matrix maptools was used for
CoppockJEPS_figure1acode.R	Fails on a missing directory; runs once it exists	Create Replication Archive/figures and .../modeloutputs; copy CoppockJEPS.RData in
CoppockJEPS_figure1bcode.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_figure2code.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_figure3code.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure2.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure3.R	Fails on a missing directory; runs once it exists	As above
CoppockJEPS_appendixfigure4.R	Fails on a missing directory; runs once it exists, and sets no seed	As above; set.seed(343), as every other script does
CoppockJEPS_appendixfigure5.R	Fails on a missing directory; runs once it exists, and its simulation never permutes	As above; draw a fresh assignment per simulation
CoppockJEPS_appendixtable.R	Fails: loads seven files by bare name and writes to a hardcoded /Users/Alex path	Run from the modeloutputs directory; redirect the output

3.1 The directory that is not in the deposit

The scripts and the deposit disagree about where files live, in both directions. Every figure script reads `Replication Archive/CoppockJEPS.rdata` and writes to `Replication Archive/figures/`; the data preparation script reads `CoppockJEPSshapefile/hd_court_ordered.shp`. Neither path exists at Dataverse, where the shapefile's three components sit at the top level alongside everything else. The published appendix documents the shapefile directory as a directory, so this is an upload that flattened a tree rather than an author who never had one.

Two smaller path defects come with it, and both are invisible on macOS.

`CoppockJEPS_10000Randomizations.R` writes `CoppockJEPS_10000randomizations.rdata`, with a lower-case r, and `CoppockJEPS_datapreparation.R` loads `CoppockJEPS_10000Randomizations.rdata`, with an upper-case one. A case-insensitive filesystem resolves that; Linux does not.

More seriously, `CoppockJEPS_datapreparation.R` ends by writing `CoppockJEPS.rdata` into the working directory, and the deposited file is `CoppockJEPS.RData`. Running the data preparation script inside a copy of the archive on macOS therefore silently overwrites a deposited file with a regenerated one. That is why everything in this report was run from a scratch copy, and why `download_original.R` reports anything in `original/` that the manifest does not list.

4 The corrigendum

The corrigendum is the substance of the case, so it is worth being precise about which object is wrong.

`CoppockJEPS.RData` stores four similarity matrices. Three of them, the rank, squared-distance and geographic ones, rebuild from the deposited ideal points and shapefile exactly. The fourth, `similarity.matrix.dw`, is 1 in every off-diagonal cell and 0 on the diagonal. It asserts that every pair of legislators in the New Mexico House is maximally and identically close in ideological space, which is precisely what the corrigendum says went wrong.

The corruption is confined to that one object. The net exposure variable the published regressions use, `s.difs.dw`, was computed from a correct similarity matrix: rebuilding it from Equation (1) reproduces the deposited values to fourteen decimal places. The published analysis therefore regressed the observed outcome on a correctly built exposure measure while simulating the null distribution with the all-ones matrix. Under that matrix every unit's raw exposure is the number of treated colleagues, which is the same for everyone, so the simulated exposure carries no variation attributable to the random assignment at all. The reference distribution is far too narrow, and the confidence regions that come out of it are far too tight. That is the mechanism behind the corrigendum.

Table 2: Figure 2 as the corrigendum states it, as the deposited scripts produce it, and as the maintained rewrite produces it. A blank archive column means the deposit contains no corrected analysis to run.

Version	Quantity	Corrigendum	Archive	Rewrite
Uncorrected	direct effect	-0.3	-0.3	-0.3
Uncorrected	direct 95% CI lower	-0.575	-0.5	-0.5
Uncorrected	direct 95% CI upper	0	-0.075	-0.075
Uncorrected	indirect effect	-0.225	-0.225	-0.225
Uncorrected	indirect 95% CI lower	-0.325	-0.325	-0.325
Uncorrected	indirect 95% CI upper	-0.025	-0.1	-0.1
Corrected	direct effect	-0.25		-0.25
Corrected	direct 95% CI lower	-0.625		-0.625
Corrected	direct 95% CI upper	0.175		
Corrected	indirect effect	-0.15		-0.15
Corrected	indirect 95% CI lower	-0.325		-0.35
Corrected	indirect 95% CI upper	0.2		0.175

Three things in that table are worth separating.

The corrigendum's account of the uncorrected result does not match the uncorrected result. It reports the original 95% confidence intervals as running from -0.325 to -0.025 for the indirect effect and from -0.575 to 0.000 for the direct effect. Appendix Table 1 of the 2014 article gives [-0.3, -0.1] and [-0.5, -0.075], and those are what the deposited scripts produce today. The point estimates agree; the interval endpoints are three grid steps apart on the direct effect. The most likely explanation is that the corrigendum re-ran the uncorrected model rather than quoting the article, and got a differently seeded surface, which is the same instability documented above. It is nonetheless a discrepancy inside the published record: two articles state different confidence intervals for the same estimate.

The correction reproduces exactly, once two things are supplied that the deposit does not carry. Rebuilding the ideological similarity matrix from the deposited ideal points and rerunning does what the corrigendum says qualitatively, widening both intervals across zero, but it returns neither the corrected point estimates nor the corrected intervals. Seeds, simulation counts, grid width and the matrix diagonal were each varied and none of them closes the gap.

The gap closes on the ideal points. The corrigendum was computed from a different set of W-NOMINATE scores than the deposit carries, and the difference traces to an unpinned package, as the Summary sets out. With those scores, and with the sampler of the day restored, the p-value surface this repository produces is identical to the one saved in the corrigendum's own working archive: all 1,369 cells, maximum difference exactly zero.

The archive nonetheless records none of that. It was released in 2015 and never revised, so a reader who downloads it today gets the retracted figures, the scores that no longer

regenerate, and no indication that either is the case. A corrigendum that corrects a published analysis is itself a published analysis, and this one was never deposited.

5 Number-by-number comparison

Table 3: Every value the corrigendum prints, against what this repository produces.

Figure	Quantity	Corrigendum	Here	Match	If not, where
figure_2 uncorrected	direct effect	-0.3	-0.3	1	
figure_2 uncorrected	direct 95% CI lower	-0.575	-0.5	0	paper_internal
figure_2 uncorrected	direct 95% CI upper	0	-0.075	0	paper_internal
figure_2 uncorrected	indirect effect	-0.225	-0.225	1	
figure_2 uncorrected	indirect 95% CI lower	-0.325	-0.325	1	
figure_2 uncorrected	indirect 95% CI upper	-0.025	-0.1	0	paper_internal
figure_2 corrected	direct effect	-0.25	-0.25	1	
figure_2 corrected	direct 95% CI lower	-0.625	-0.625	1	
figure_2 corrected	direct 95% CI upper	0.175			
figure_2 corrected	indirect effect	-0.15	-0.15	1	
figure_2 corrected	indirect 95% CI lower	-0.325	-0.35	0	environment
figure_2 corrected	indirect 95% CI upper	0.2	0.175	0	environment

The corrigendum prints two sets of values: the corrected estimates it substitutes, and a restatement of the original ones it is correcting. Both are recorded.

Of the corrected set, 3 of 5 return exactly. Of the restatement, three of six disagree with the article they restate. The corrigendum gives the original direct interval as -0.575 to 0.000, where the 2014 appendix gives a narrower one, and the archive produces the appendix's. Two published documents state different intervals for the same estimate, and this repository reproduces the earlier one.

6 Maintained rewrite

The rewrite lives in `maintained/`: one cleaning script, nine figure scripts, two table scripts and one script for the in-text quantities.

6.1 Architecture

`clean_data.R` is the load-bearing one. It reads the deposited `CoppockJEPS.RData`, rebuilds all four similarity matrices from the deposited ideal points and shapefile, and prints how far each rebuilt matrix sits from the deposited one, which is what localizes the defect: three deviations of zero and one of 1. It then computes expected and observed net exposures for each matrix,

and keeps the deposited all-ones matrix alongside the corrected one so that the uncorrected figures can be reproduced rather than described.

Every spillover figure is therefore computed twice, once from each matrix, and both versions go into `output/`. Nothing is transcribed: the corrigendum's numbers appear in the ground truth as comparison targets and nowhere in the code.

The p-value grid itself is one function, `pvalue_grid()` in `helpers.R`, taking the similarity matrix, the expected exposures and the observed exposure as separate arguments. That separation is deliberate. The difference between the published analysis and the corrected one is exactly which matrix goes into the simulation while the observed regressor stays the same, and a helper that fixed either of them internally would hide the only choice in the paper that matters.

The rewrite's p-value matrices were checked cell by cell against the ones the deposited scripts save. Six of the seven are bit-identical: Figure 1a's 71 cells, Figure 1b's 2,205, Figure 2's 1,369, Figure 3's 1,305, Appendix Figure 2's 1,681 and Appendix Figure 3's 2,401, with a maximum absolute deviation of exactly zero in every case. Appendix Figure 4 is the exception, and only because its script sets no seed. Agreement at that level is what makes the point-estimate differences interpretable: the surfaces are the same, so any difference in what is read off them is a difference in the reading.

The test statistic goes through `.lm.fit` rather than `lm`. The rewrite fits about eleven million regressions; `lm` takes 0.295 milliseconds each and `.lm.fit` takes 0.0035, and the two agree on the sum of squared residuals to floating point. That is the difference between a five-minute run and an hour-long one, and a reproduction check nobody runs is not a check.

6.2 Deprecated patterns replaced

Table 4: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd("")‘ and ‘Replication Archive/‘ paths	‘here::here()‘
‘maptools::readShapeSpatial‘	‘sf::st_read‘ + ‘spdep::poly2nb‘
‘lattice::levelplot‘ + ‘larrows‘	‘ggplot2::geom_tile‘ + ‘annotate(”segment”)‘
‘pdf()‘ / ‘dev.off()‘	‘ggsave()‘, PDF and PNG
‘xtable‘ + ‘sink()‘ to a hardcoded path	‘write_csv()‘ to ‘output/‘
‘save()‘ / ‘load()‘ of .rdata	‘write_rds()‘ / ‘read_rds()‘
nested ‘for‘ loops over ‘data.frame(...)‘ + ‘lm()‘	design matrices + ‘.lm.fit‘
‘Z_block[, 1]‘ inside the simulation loop	‘z_block[, sample(ncol(z_block), 1)]‘

6.3 Summary tables

Table 5: Appendix Table 1, constant effects models, as the maintained rewrite produces it.

	No spillover	Absolute distance	Absolute distance (uncorrected)	Rank distance	Squared distance	Geographic
Direct Effect	-0.16	-0.25	-0.3	-0.225	-0.25	-0.15
Direct 95% CI	[-0.34,0.02]	[-0.625,NA]	[-0.5,-0.075]	[-0.45,0.1]	[-0.525,0.325]	[-0.37,0.09]
Indirect Effect		-0.15	-0.225	-0.2	-0.175	0.025
Indirect 95% CI		[-0.35,0.175]	[-0.325,-0.1]	[-0.325,0.15]	[-0.325,0.525]	[-0.095,0.15]
Corresponding Figure	1a	2	2	A2	A3	A4

Table 6: Appendix Table 2, heterogeneous effects models, as the maintained rewrite produces it.

	No spillover	Absolute distance	Absolute distance (uncorrected)
Direct Effect, High Support	0.05	-0.125	-0.1875
Direct 95% CI, High Support	[-0.225,0.35]	[-0.65,NA]	[-0.4,0.075]
Indirect Effect, High Support		-0.225	-0.3
Indirect 95% CI, High Support		[-0.45,NA]	[-0.425,-0.175]
Direct Effect, Low Support	-0.375	-0.425	-0.475
Direct 95% CI, Low Support	[-0.675,-0.05]	[-0.775,0.05]	[-0.8,-0.125]
Indirect Effect, Low Support		-0.1	-0.175
Indirect 95% CI, Low Support		[-0.325,0.15]	[-0.325,0]
Corresponding Figure	1b	3	3

6.4 In-text quantities

Table 7: The page 167 arithmetic, under both similarity matrices.

Quantity	Value
Legislators treated	35
Difference-in-means estimate (Figure 1a)	-0.16
Implied change in votes under the difference in means	-5.6
Legislators voting yea	51
Yea votes in the uniformity trial, uncorrected	61.46
Total causal effect in votes, uncorrected	10.46
Yea votes in the uniformity trial, corrected	59.64
Total causal effect in votes, corrected	8.64

The article states that 61 legislators would have voted yea in the uniformity trial against 51 who actually did, for a total causal effect of 10 votes, roughly twice the 5.6 the difference in means implies. The uncorrected column reproduces that. The corrected column is the quantity the corrigendum's conclusion bears on: the point estimates barely move, so the arithmetic barely moves, and what changes is that neither effect can be distinguished from zero.

7 Figure verification

The correction's visible consequence is the width of the shaded region. The uncorrected map is a tight blob well away from both axes; the corrected map spreads across zero on both.

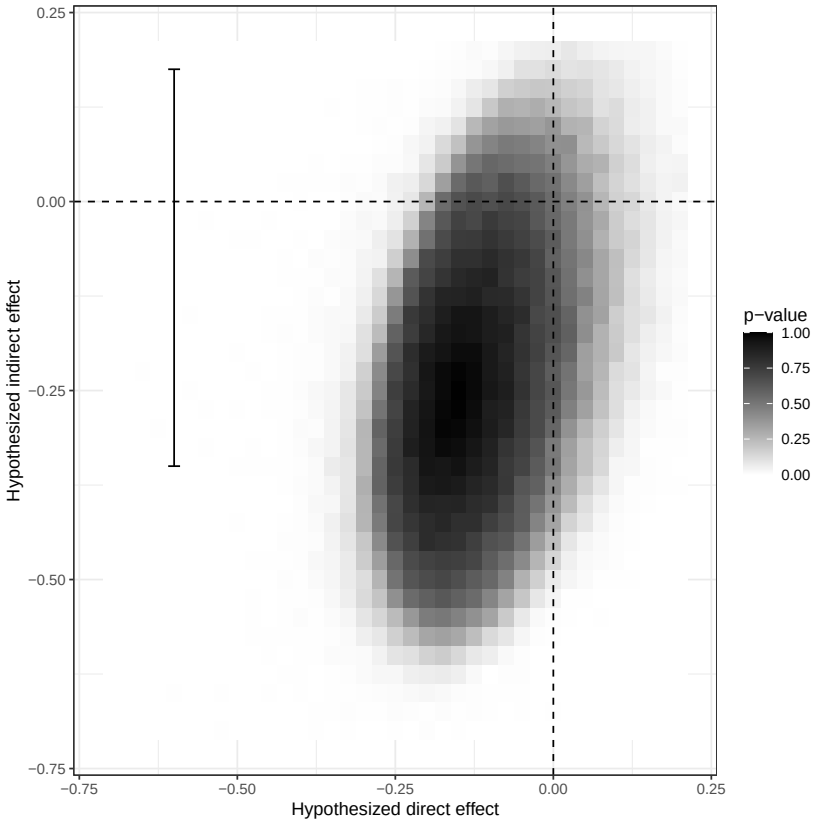


Figure 1: Figure 2, corrected similarity matrix.

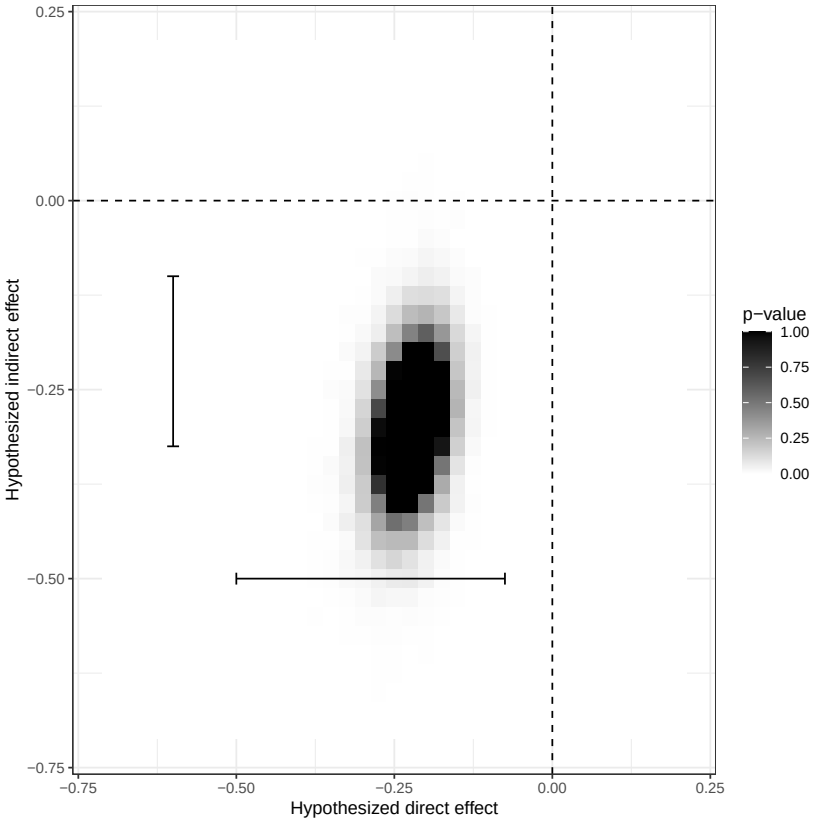


Figure 2: Figure 2 as published, all-ones similarity matrix.

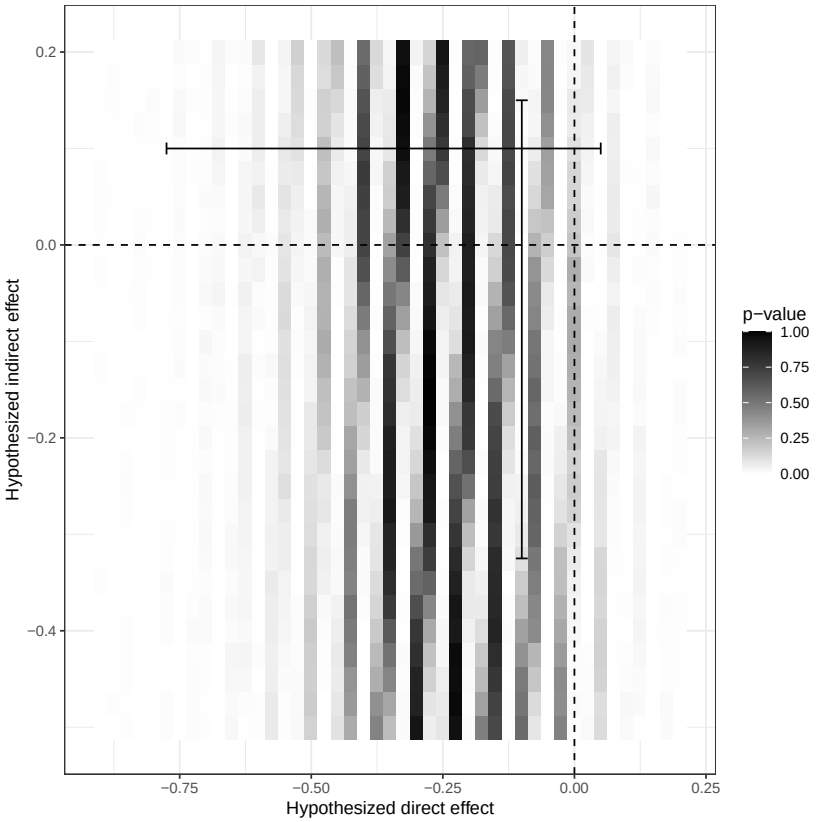


Figure 3: Figure 3a, lower-support districts, corrected.

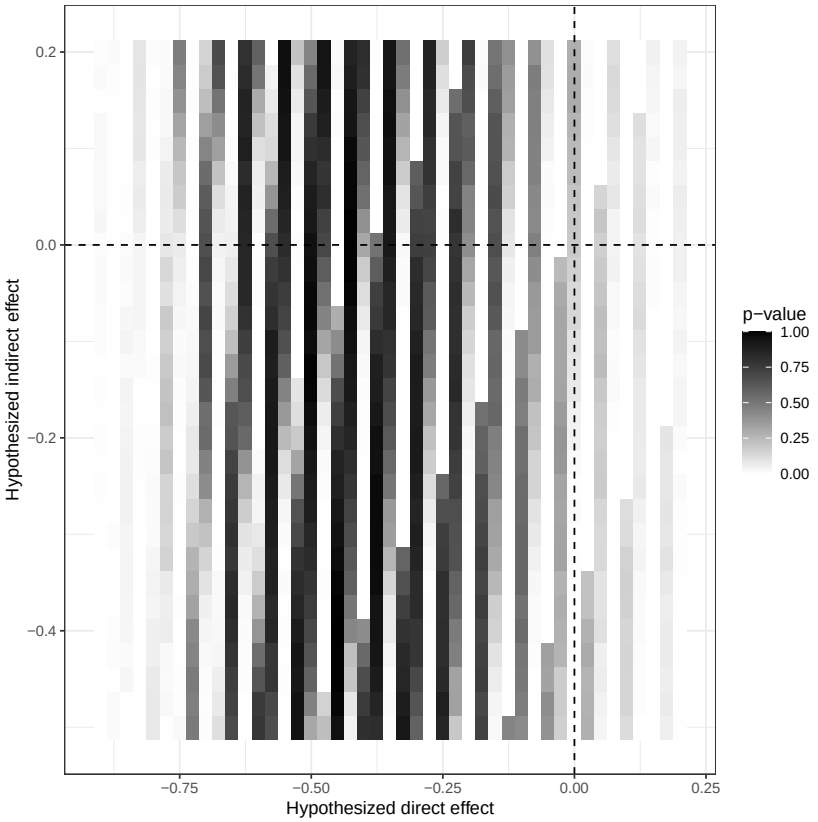


Figure 4: Figure 3b, higher-support districts, corrected.

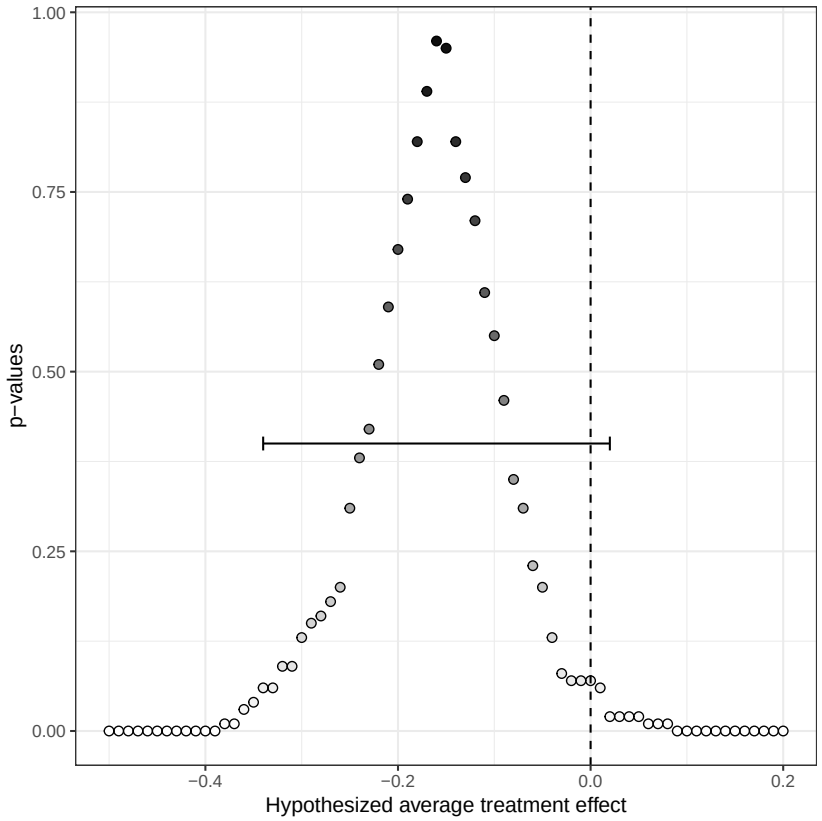


Figure 5: Figure 1a, the no-spillover constant effects model, which reproduces exactly.

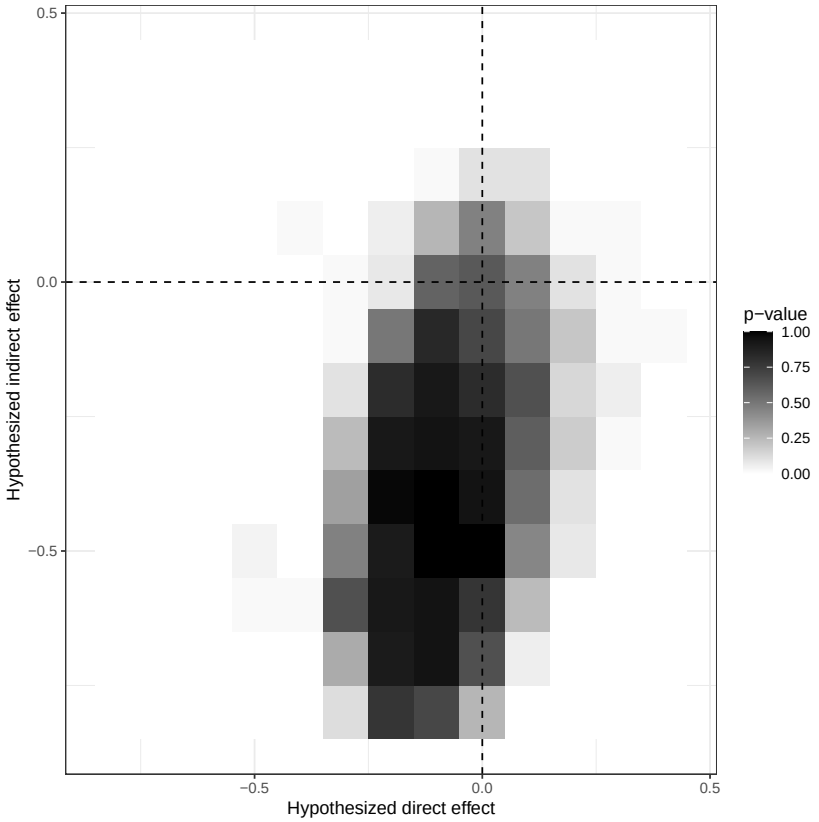


Figure 6: Appendix Figure 5a with a fresh assignment drawn per simulation.

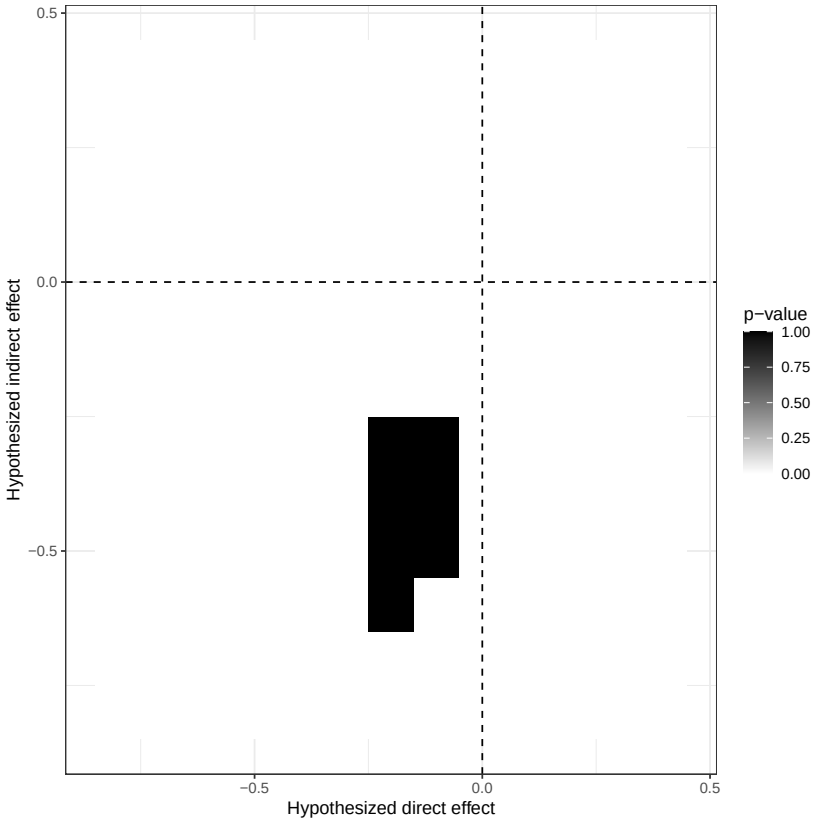


Figure 7: Appendix Figure 5a as published: every p-value is 0 or 1.

8 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23
Date run	2026-08-01

Table 9: Package versions used for the run behind this report.

Package	Version
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2

purrr	1.2.2
readr	2.2.0
sf	1.1.2
spdep	1.4.2
here	1.0.2